PharmAchieve

HYPERTHYROIDISM

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

Providing Care:
Clinical Care

Providing Care:
Distribution

Knowledge
& Expertise

Communication
& Collaboration

Leadership &
Stewardship

Professionalism

LEGEND

ATD	Antithyroid Drug
C/I	Contraindication
COPD	Chronic Obstructive Pulmonary Disease
D/C	Discontinue
D/I	Drug Interaction
(f)T3	(Free) Triiodothyronine
(f)T4	(Free) Thyroxine
INR	International Normalized Ratio
LFT	Liver Function Test
MMI	Methimazole
MOA	Mechanism Of Action
NSAID	Non-Steroidal Anti-inflammatory Drug
N/V/D	Nausea/Vomiting/Diarrhea
PTU	Propylthiouracil
PVD	Peripheral Vascular Disease
RAI(U)	Radioactive Iodine (Uptake)
TKI	Tyrosine Kinase Inhibitor
TRAb	Thyrotropin Receptor Antibodies
TSH	Thyroid-Stimulating Hormone

PATHOPHYSIOLOGY

Graves' Disease

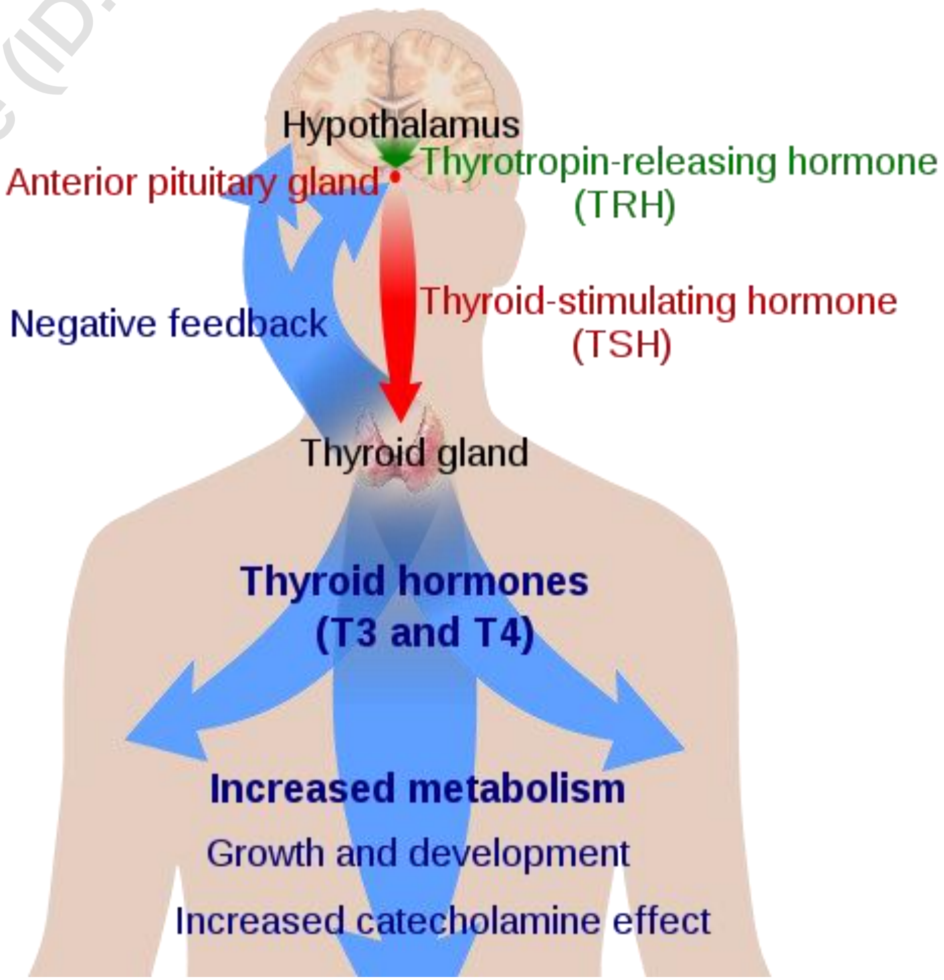
- Most common cause
- Thyroid receptor stimulating immunoglobulins (TRAb) stimulate TSH receptor on thyroid gland
- Overproduction of thyroid hormone
- Smoking is a risk factor

Toxic Nodule or Multi-Nodular Goitre

- Thyroid growth and activity of nodules cause increase in overall hormone production

Subacute (Viral)/Acute (Bacterial) Thyroiditis

- Infection of thyroid gland
- Leakage of thyroid hormone



CLINICAL PRESENTATION

- Nervousness
- Perspiration
- Palpitations
- Hand Tremors
- Anxiety
- Diarrhea
- Heat Intolerance
- Weight loss
- Insomnia

- Increased appetite
- Eyelid lag
- Tachycardia
- Hyperreflexia
- Warm/moist skin
- Goitre
- Nodules
- Increased risk of atrial fibrillation

Increased metabolic activity

COMPLICATIONS

Graves' Ophthalmopathy (Exophthalmos)

↑ in extraocular muscle volume and retro-orbital connective and adipose tissues
↑ fluid content and pressure in the eye: Forward eye displacement

Thyroid Storm

- Life-threatening with severe thyrotoxicosis (heart rate, blood pressure and body temperature can reach dangerously high levels)
- Excessive thyroid hormone production
- Causes: radioactive iodine, infection, trauma, surgery, discontinuation of anti-thyroid agents
- Treat with propylthiouracil, beta blockers, corticosteroids, Lugol's solution (iodine to block thyroid hormone production)
- NSAIDs can worsen thyrotoxicosis by displacing thyroid hormones from globulins. Use acetaminophen instead to treat high temperatures

DRUG CAUSES

Levothyroxine, Liothyronine

Over-replacement of thyroid hormone

Amiodarone

May cause excessive free iodine load

Lithium (rare)

Interferon-α

Tyrosine Kinase Inhibitors (TKIs)

DIAGNOSIS

Condition	Laboratory Values
Graves' Disease	• ↑RAIU, ↓ TSH, ↑ fT_4 • Diffuse pattern on thyroid scan • Subclinical hyperthyroidism: Normal fT_4, ↓TSH
Toxic Nodule	• 0% or ↑ RAIU, ↓ TSH, ↑ fT_4 • Hot area on thyroid scan
Toxic Multi-nodular Goitre	• ↑RAIU, ↓ TSH, ↑ fT_4 • Multiple hot areas on thyroid scan
Thyroiditis	• 0% RAIU, ↓ TSH, ↑ fT_4 • Gland is poorly defined on thyroid scan

Diagnosis based on:

1. Symptoms
2. TSH
3. fT_3 / fT_4
4. RAIU
5. Thyroid scans

Note: Biotin can interfere with thyroid tests. Patients taking biotin supplements should discontinue them 48 hours prior to the test.

GOALS OF THERAPY

- ✓ Achieve and maintain a euthyroid state
- ✓ Prevent hyperthyroidism-related complications
- ✓ Manage the signs and symptoms of hyperthyroidism
- ✓ Prevent drug-related adverse effects

NON-PHARMACOLOGICAL MEASURES

- Thyroid surgery for thyroid nodules, large goiters, and thyroid cancer
- Medical therapy usually used prior to surgery to create euthyroid state

PHARMACOLOGICAL MEASURES - OVERVIEW

<table>
<tr><th>Drug Class: MOA</th><th>Adverse Effects/Safety</th><th>D/I</th></tr>
<tr><td>Radioactive Iodine ^{131}I:
• Radiation damages thyroid tissue</td><td>• N/V/D, metallic taste, sore neck, radiation thyroiditis, hypothyroidism
• Hypersensitivity reactions, blood dyscrasias
• C/I: Pregnancy/breastfeeding
• Worsens Graves' ophthalmopathy</td><td>• Amiodarone, levothyroxine
• Anti-thyroid agents decrease effect: D/C 2-3 days prior
• Lithium prolongs ^{131}RAI activity</td></tr>
<tr><td>Methimazole:
• Inhibits incorporation of iodine to tyrosine residues</td><td rowspan="2">• Arthralgia, rash, nausea can improve in 4 weeks. Divided doses can help nausea
• Rare: edema, bone marrow suppression, agranulocytosis, hepatic/autoimmune effects
• C/I: methimazole in 1st trimester of pregnancy
• Higher risk of hepatotoxicity and agranulocytosis with propylthiouracil in first 3 months (first trimester)
• Patients should stop taking ATDs if they experience fever, jaundice, or a rash during therapy</td><td rowspan="2">• Clozapine (agranulocytosis)
• Digoxin: May require digoxin dose reduction once euthyroid
• Warfarin: May affect INR</td></tr>
<tr><td>Propylthiouracil:
• Same as methimazole + inhibits peripheral T4 to T3 conversion</td></tr>
</table>

PHARMACOLOGICAL MEASURES - TIPS

Radioactive Iodine 131-I	• After therapy, wait at least 6 months before conceiving • Women of childbearing age require a negative pregnancy test within 48 hours of receiving radioactive iodine • Radioiodine should not be given for 6 to 12 weeks after cessation of breastfeeding
Methimazole (MMI)	• Preferred over PTU: faster reversal of hyperthyroidism, fewer side effects, longer duration of action • Can be used in children Treatment with thionamides (methimazole or PTU) may be life-long • Often treat until euthyroid for at least 12-18 months • If discontinuing, TSH, fT3 and fT4 should be monitored every 2-3 months for 6 months, then 6 months after this, followed by every 6-12 months. If euthyroid for a year, can monitor annually • Relapse varies (days to years) based on presence of TRAb
Propylthiouracil (PTU)	• Indication: minor reactions to MMI who do not want definitive treatment with surgery/RAI (50% cross-sensitivity) • Preferred in life-threatening thyrotoxicosis since prevents conversion of T4 to T3, and 1st trimester of pregnancy due to lower risk of congenital abnormalities • Avoid in children (high risk of hepatotoxicity) • Can cause thyroid to be more resistant to 131-I

PHARMACOLOGICAL MEASURES - TIPS

Beta-Blockers	• Recommended in all patients with symptoms of hyperthyroidism, especially in the elderly, those with a resting heart rate of > 90 beats per minute, or those with concomitant cardiovascular disease • Controls heart rate, improves daily functioning and fatigue • Propranolol has most evidence for use. Doses over 160 mg/day may block conversion of T4 to T3 • Avoid using non-selective beta-blockers (propranolol) in patients with confirmed asthma/COPD, PVD, Raynaud's → use cardio-selective beta-blockers instead (atenolol, bisoprolol, metoprolol) • Adverse effects include bradycardia, hypotension, hyperglycemia, bronchospasm, and PVD exacerbation

SPECIAL POPULATIONS - PREGNANCY AND BREASTFEEDING

- PTU is preferred in first-trimester of pregnancy
- Switch to methimazole in second trimester and continue during breastfeeding
- Should achieve good control of thyroid levels prior to conception
- Increased risk of fetal loss with untreated hyperthyroidism
- May ameliorate during the second and third trimester
- May flare again in the postpartum period
- Newborn should be assessed for hypothyroidism if mother is on a MMI or PTU
- Surgery should be avoided in the first trimester. However, significant risks are still possible with surgery in the second and third trimesters.
- Avoid atenolol use in pregnancy due to the risk of smaller babies
    - Beta-blockers should be used for the shortest duration and at the lowest dose if needed

MONITORING PARAMETERS

EFFICACY ENDPOINTS	
Parameter and Degree of Change	**Timeframe**
TSH: 0.34-5.60 mIU/L Free T4: 7.0-17.0 pmol/L Free T3: 3.3-6.0 pmol/L	Measure every 4-6 weeks until euthyroid Then monitor every 2-3 months If >18 months therapy, monitor every 6 months
Improvement in symptoms of hyperthyroidism	2-3 weeks Peak effect may take 4-6 weeks

SAFETY ENDPOINTS	
Parameter and Degree of Change	**Timeframe**
Prevent symptoms of hypothyroidism	Duration of therapy If identified, treat as per hypothyroidism
Prevent neutropenia Prevent LFT > 5x normal	Duration of therapy Obtain baseline values and monitor

TAKEAWAY

- MMI preferred over PTU; faster, lasts longer, less side effects, once daily dosing
- PTU preferred over MMI in life-threatening thyrotoxicosis, 1st trimester of pregnancy and in patients with minor reactions to MMI who do not want definitive treatment
- Discontinue anti-thyroid drugs 2-3 days prior to RAI as they can decrease its effect
- Monitor thyroid blood tests every 4-6 weeks until stable

CASE 1

GD is a 29-year-old male, working in construction. He has been feeling more nervous and agitated recently. He has loose bowel movements for the last few weeks. The guys at work have commented that he has lost weight, but he has been eating normally. He finds himself feeling more hot and clammy than usual. Last week on the site, he found it difficult to use his tools because his hands were trembling. He also noticed his heart racing often. He decided to see a doctor who initiated atenolol 25 mg daily. He has not had any improvement in symptoms despite starting atenolol a month ago. He does not take any other medications. His past medical history is significant for benign postural vertigo. His father had Grave's Disease, his mother had diabetes, his maternal grandmother had diabetes and his paternal grandmother had dementia. He is not allergic to anything. He smokes 25 cigarettes/day for the last 14 years, drinks 6 alcoholic drinks per week and uses cannabis 3-4 times per week.

Objective findings:

- TSH < 0.01 (0.34 – 5.6 mIU/L)
- fT4 = 31 (7.0 – 17.0 pmol/L)
- fT3 = 19.3 (3.3 – 6.0 pmol/L)
- BP = 161/77 mmHg, HR = 84 bpm

1. Which of GD's presenting symptoms and risk factors are consistent with hyperthyroidism?
   a) Heat intolerance, weight loss, tachycardia, smoking, familial history
   b) Heat intolerance, weight loss, bradycardia, smoking, familial history
   c) Heat intolerance, weight loss, tachycardia, alcohol consumption, familial history
   d) Heat intolerance, increased appetite, tachycardia, cannabis use, familial history

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2. What is the next best option for treatment for GD?

   a) Increase dose of atenolol to 50 mg once daily
   b) Initiate methimazole
   c) Initiate propylthiouracil
   d) A and C

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3. GD decides to pursue pharmacotherapy. His response to therapy should be assessed via TSH in:

   a) 1 month
   b) 3 months
   c) 12 months
   d) 18 months

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4. Based on antithyroid agent selected in question 2, all of the following are appropriate counselling points EXCEPT:

    a) You do not need to skip doses of methimazole before certain medical tests/procedures
    b) You may be able to stop this medication in 12 - 18 months
    c) Quitting smoking can help improve your condition
    d) If you develop yellowing of the skin/eyes, fever or sore throat, it's important to quickly seek medical attention

CASE 2

PT is a 34-year-old female who is 4 weeks pregnant with her second child.

PT has had a relapsing-remitting headache for 2 weeks. She reports feeling much more nauseous than she did with her first pregnancy. To her surprise, she has lost about 5 lbs since conceiving. She also has been feeling nervous for many days without an identified cause, which is unusual for her. Her past medical history is significant for an adenoidectomy, breast augmentation and a C-section. She takes Diclectin® (doxylamine 10 mg and pyridoxine 10 mg) 2 tablets at bedtime, and sumatriptan 50 mg as a single dose PRN as well as a prenatal vitamin 1 tablet twice daily. She is allergic to contrast media. She is currently studying business and does not smoke or drink alcohol.

Objective findings:

- TSH = 0.18 (0.34 – 5.6 mIU/L)
- Free T3 = 13 (3.3 – 6.0 pmol/L)

1. PT has been diagnosed with hyperthyroidism. Which of the following is an optimal treatment option for PT currently?

   a) Methimazole
   b) Propylthiouracil
   c) Propranolol
   d) No treatment should be initiated at this time

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Objective findings:

- TSH = 0.18 (0.34 – 5.6 mIU/L)
- Free T3 = 13 (3.3 – 6.0 pmol/L)

2. All the following are appropriate goals of therapy for hyperthyroidism in pregnancy, EXCEPT:

   a) Achieve fT3 and fT4 levels near the upper limits of normal
   b) Prevent fetal loss
   c) Avoid drug-related congenital abnormalities
   d) TSH should be below the limits of normal at conception

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3. What would be the ideal drug of choice for PT in her second trimester?
    a) Methimazole
    b) Propylthiouracil
    c) Methimazole and propranolol
    d) Propylthiouracil and propranolol

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4. All the following are appropriate counselling points EXCEPT:

   a) The medication will not affect the baby and monitoring is not required
   b) Seek medical attention if she notices yellowing of the skin/eyes, fever or sore throat
   c) Reassure her that risk of untreated hyperthyroidism is greater than risk of medications used to treat it
   d) Hyperthyroidism may resolve in 2nd or 3rd trimester, obliterating the need for medication

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CHANGE LOG

July 2025

- Content reviewed, no changes made

October 2025

- Slides 15, 17-22: Edited reference ranges for TSH, FT3 and FT4 to align with the Medical Council of Canada list of normal lab values
- Slide 25: Updated references

January 2026

- Slide 24: Removed bonus question because there is some evidence that sumatriptan is relatively safe in pregnancy, although non-pharmacological and other pharmacological options should be trialled first

February 2026

- Slide 2: Updated NAPRA competencies